Gas Laws

1. (c) Using Charle's law $\frac{P_1}{P_2} = \frac{T_1}{T_2}$

or $P_2 = \frac{P_1 T_2}{T_1} = \frac{P(273+927)}{(273+27)} = 4P$.

2. (b) $\frac{V_1}{V_2} = \frac{T_1}{T_2} \Rightarrow T_2 = 2 \times T_1 = 2 \times (273 + 0) = 546K$

$\Rightarrow T_2 = 273 \times 2 = 546K \Rightarrow 273°C \Rightarrow 273°C$

3. (a) At low pressure and high temperature, the molecules of real gases are farther apart so that molecular size is negligible as compared to the size of the vessel and also molecular force donot come in.

4. (b) In isothermal changes, temperature remains constant.

5. (c) S.I. unit of *R* is *J/mol-K.*

6. (c) Molecules of ideal gas behaves like perfectly elastic rigid sphere.

7. (c) $V \propto T \Rightarrow \frac{V_1}{V_2} = \frac{T_1}{T_2} \Rightarrow \frac{V}{3V} = \frac{(273+27)}{T_2}$

$\Rightarrow T_2 = 900K \rightarrow 627°C$ [$T(K) = 273 + t°C$]

8. (c) At constant pressure, ρT = constant

$\Rightarrow \frac{\rho_1}{\rho_2} = \frac{T_2}{T_1} \Rightarrow \frac{24}{\rho_2} = \frac{(273+127)}{(273+27)} = \frac{400}{300} \Rightarrow \rho_2 = 18$.

9. (a) $V \propto T \Rightarrow \frac{V_1}{V_2} = \frac{T_1}{T_2} \Rightarrow \frac{200}{V_2} = \frac{(273+20)}{(273-20)} = \frac{293}{253}$

$\Rightarrow V_2 = \frac{200 \times 253}{293} = 172.6ml$

10. (d) Thermal equilibrium implies that the temperature of gases is same. Hence Boyle's law is applicable *i.e.*

$P_a V_a = P_b V_b$

11. (a) $PV = \mu RT,\ T = 0,\ \therefore P = 0$ because $V \neq 0$

12. (d) Helium is much nearer to the ideal behavior due to octet in its outer shell (due to monoatomicity).

13. (b) From Andrews curve.

14. (d) From Andrews curve.

15. (c) At triple point, boiling and freezing point are same.

16. (b) Boyle's law is applicable only when the mass and temperature of ideal gas are constant.

17. (c) Mean free path $\lambda = \frac{1}{\sqrt{2}\pi d^2 n} \Rightarrow \lambda \propto d^{-2}$

18. (a) Boyle's law states that at constant temperature, pressure of given mass of a gas is inversely proportional to its volume. This law is applicable for perfect gas only.

19. (a) Closed vessel *i.e.*, volume is constant

$$\Rightarrow \frac{P_1}{P_2} = \frac{T_1}{T_2} \Rightarrow \frac{P}{P + \left(\frac{0.4}{100}\right)P} = \frac{T}{T+1} \Rightarrow T = 250K$$

20. (d) Mass of 1 *cc* of gas at sea level and at a given height does not remain the same and gas laws hold only for a constant.